

KlausurPrep

Access Modifiers

- Public: von überall zugreifbar
- Protected: von innerhalb desselben package als auch den Unterklassen
- Default: von innerhalb desselben package zugreifbar
- Private: nur von innerhalb der Klasse zugreifbar

Begriffe und Definitionen

- Fachbegriffe in den Vorlesungen
 - Java Keywords (extends, implements, static, final, private, ...)
 - Begriffe und Definitionen auf Spickzettel vorbereiten
- mehr Zeit für die Nachdenk Aufgaben

Design Patterns

- <https://refactoring.guru/design-patterns/catalog>
- kurze Erklärung + Vor- & Nachteile
- Schaut euch die Design Patterns in den Hausaufgaben an: Singleton, Builder, Factory, Observer, Strategy

List

add(int index, E element): Inserts an element at a specific position, shifting subsequent elements to the right.

remove(int index): Removes the element at the specified position.

size(): Returns the number of elements.

isEmpty(): Returns **true** if the collection contains no elements.

contains(Object o): Returns **true** if the collection contains the specified element.

get(int index): Retrieves the element at the specified position.

set(int index, E element): Replaces the element at the specified position with a new element.

Map

put(K key, V value): Associates the value with the key. If the key already exists, the old value is replaced.

get(Object key): Returns the value associated with the key, or **null** if the key doesn't exist.

remove(Object key): Removes the mapping for a key.

containsKey(Object key) / containsValue(Object value): Checks if a specific key or value exists.

keySet(): Returns a **Set** of all the keys.

values(): Returns a **Collection** of all the values.

entrySet(): Returns a **Set** of **Map.Entry<K, V>** objects (the best way to iterate over both keys and values simultaneously).